

Week 3

- **Problem 1**

We say that two integers have *different parity* if one of them is odd and the other one is even. Prove that two integers have different parity if and only if their sum is odd.

- **Proof 1.1**

($\Rightarrow$) Suppose two integers have different parity. Without loss of generality, let one of them be $2k_1$ for some $k_1 \in \mathbb{Z}$ and the other one be $2k_2 + 1$ for some $k_2 \in \mathbb{Z}$. Then their sum

$$2k_1 + (2k_2 + 1) = 2(k_1 + k_2) + 1$$

is odd because $k_1 + k_2 \in \mathbb{Z}$.

($\Leftarrow$) Let the two integers be a and b , and suppose that their sum is odd, that is, $a + b = 2s + 1$ for some $s \in \mathbb{Z}$. By definition, the integer a must be either even or odd. We consider these two cases:

1. **Case 1:** Suppose a is even, so $a = 2k_1$ for some $k_1 \in \mathbb{Z}$.

Then we can express b as:

$$b = (a + b) - a = (2s + 1) - 2k_1 = 2(s - k_1) + 1.$$

Since $s - k_1 \in \mathbb{Z}$, b is an odd integer. Thus, a is even and b is odd, meaning they have different parity.

2. **Case 2:** Suppose a is odd, so $a = 2k_1 + 1$ for some $k_1 \in \mathbb{Z}$.

Then we can express b as:

$$b = (a + b) - a = (2s + 1) - (2k_1 + 1) = 2(s - k_1).$$

Since $s - k_1 \in \mathbb{Z}$, b is an even integer. Thus, a is odd and b is even, meaning they have different parity.

Therefore, in either case, the two integers must have different parity.

- **Problem 2**

Suppose that n lines are drawn in the plane so that

1. no two lines are parallel;
2. no three lines pass through the same point.

Prove by induction that the lines divide the plane into

$$1 + \frac{n(n+1)}{2}$$

regions.

- **Proof 2.1**

By induction.

1. Base step

If $n = 1$ the result is true since the number of regions is $1 + \frac{1 \cdot (1+1)}{2} = 2$.

2. Inductive step

Suppose that the result is true for n , that is, any n lines satisfying the conditions divide the plane into

$$1 + \frac{n(n+1)}{2}$$

regions.

Now consider the case with $n + 1$ lines. When we add the $(n + 1)$ -th line to the existing n lines:

- Since no two lines are parallel and no three lines pass through the same point, the new line must intersect all n existing lines at exactly n distinct, non-overlapping intersection points.
- These n distinct intersection points divide the new $(n + 1)$ -th line into exactly $n + 1$ line segments or rays at the ends.
- Each of these $n + 1$ segments cuts through an existing region and splits it into two, thereby creating exactly $n + 1$ new regions.

Hence, the total number of regions for $n + 1$ lines is

$$\begin{aligned} \left(1 + \frac{n(n+1)}{2}\right) + (n+1) &= 1 + \frac{n(n+1) + 2(n+1)}{2} \\ &= 1 + \frac{(n+1)(n+2)}{2} \\ &= 1 + \frac{(n+1)((n+1)+1)}{2}. \end{aligned}$$

Then the result is true for $n + 1$ and therefore the lines divide the plane into $1 + \frac{n(n+1)}{2}$ regions for every $n \in \mathbb{N}$.

• Problem 3

Prove by induction that for every $n \in \mathbb{N}_0$,

$$4 \mid (5^{n+1} + 3 \cdot 9^n - 4).$$

◦ Proof 3.1

By induction.

1. Base step

If $n = 0$ then the result is true because $5^{0+1} + 3 \cdot 9^0 - 4 = 4$ and $4 \mid 4$.

2. Inductive step

Suppose that the result is true for n , that is,

$$4 \mid (5^{n+1} + 3 \cdot 9^n - 4).$$

By definition, $\exists k \in \mathbb{Z}, 5^{n+1} + 3 \cdot 9^n - 4 = 4k \implies 5^{n+1} + 3 \cdot 9^n = 4(k+1)$. Then

$$\begin{aligned} 5^{(n+1)+1} + 3 \cdot 9^{n+1} - 4 &= 5 \cdot 5^{n+1} + 9 \cdot 3 \cdot 9^n - 4 \\ &= 5 \cdot (5^{n+1} + 3 \cdot 9^n) + 4 \cdot 3 \cdot 9^n - 4 \\ &= 5 \cdot 4(k+1) + 4 \cdot 3 \cdot 9^n - 4 \\ &= 4 \cdot (5k + 4 + 3 \cdot 9^n). \end{aligned}$$

Since $5k + 4 + 3 \cdot 9^n \in \mathbb{Z}$, we have $4 \mid (5^{(n+1)+1} + 3 \cdot 9^{n+1} - 4)$. Then the result is true for $n + 1$ and therefore

$$4 \mid (5^{n+1} + 3 \cdot 9^n - 4)$$

is true for every $n \in \mathbb{N}_0$.

• Problem 4

Suppose $a, c \in \mathbb{N}$ with

$$a < c \quad \text{and} \quad a \mid c.$$

Suppose also that

$$c - a \mid c + a.$$

Determine all possible values of c in terms of a . Justify your answer.

Your proof should use only the definition and basic properties of divisibility.

◦ **Solution 4.1**

Suppose $a, c \in \mathbb{N}$ with $a < c$, $a \mid c$, $c - a \mid c + a$. The only possible values of c in terms of a are $c = 2a$ and $c = 3a$.

By the basic properties of divisibility, since $c - a \mid c + a$ and $c - a \mid c - a$, it must divide their difference:

$$c - a \mid (c + a) - (c - a) \implies c - a \mid 2a.$$

Since $a \mid c$ and $a < c$, by the definition of divisibility, there exists an integer $k \in \mathbb{N}$ with $k > 1$ such that

$$c = ka.$$

Substituting $c = ka$ into the relation $c - a \mid 2a$, we get

$$ka - a \mid 2a \implies a(k - 1) \mid 2a.$$

Since $a \in \mathbb{N}$, by the basic properties of divisibility

$$k - 1 \mid 2.$$

Since $k > 1$, the term $k - 1$ is a positive integer. The only positive divisors of 2 are 1 and 2. This gives two possible cases: $k = 2$ or $k = 3$, that is, $c = 2a$ or $c = 3a$.

• **Problem 5**

Suppose $x, y \in \mathbb{N}$ with

$$x^6 = 81y^{10}.$$

Prove that every prime that divides y also divides x .

◦ **Proof 5.1**

Suppose $x, y \in \mathbb{N}$ with $x^6 = 81y^{10}$. Let p be a prime number such that $p \mid y$. Since $y \in \mathbb{N}$, by definition, we have $\nu_p(y) \geq 1$.

By Proposition, we have

$$\begin{aligned} \nu_p(x^6) &= \nu_p(81y^{10}) \\ 6\nu_p(x) &= \nu_p(81) + 10\nu_p(y). \end{aligned}$$

Since $81 = 3^4$, we know that $\nu_p(81) \geq 0$ for any prime p . Together with $\nu_p(y) \geq 1$, we obtain that

$$6\nu_p(x) = \nu_p(81) + 10\nu_p(y) \geq 0 + 10(1) = 10.$$

This implies $6\nu_p(x) \geq 10$, which means $\nu_p(x) \geq \frac{10}{6} > 0$.

Since $\nu_p(x) \in \mathbb{N}_0$, it must be that $\nu_p(x) \geq 1$. By definition, this means $p \mid x$.

Therefore, every prime that divides y also divides x .